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Explanation-Aware Incentives for Reducing Strategic Manipulation in Multi-Agent AI: A Cross-Domain Evaluation with Blockchain Provenance

Abstract When several AI agents decide together, each with its own objective, an agent can steer the group by voting for the option it privately prefers and attaching a misleading explanation, and ordinary payoffs neither reward faithful explanations nor penalise false ones. We propose an explanation-aware incentive: an auditor scores each agent's report for fidelity to the SHAP attributions of the agent's own model, for evidence supported by the observed data, and for manipulation, and these scores enter the agent's utility; rewards accumulate into a reputation that weights votes, and a blockchain records hashes of decisions, explanations, evidence, rewards and reputation. Under a pre-registered protocol, five learning agents per domain were evaluated on two real datasets, USAID/PEPFAR pharmaceutical deliveries (supplier selection) and an energy community with peer-to-peer trading, with ten paired seeds. The incentive reduced manipulation in both domains (*-69.7%* and *-44.4%*). It improved explanation fidelity and evidence support strongly where manipulation paid under the baseline (pharmaceutical: *+83.5%*, *+75.7%*) and only negligibly where agents were already largely honest (energy). It did not make collective decisions more robust to a growing share of agents that always manipulate. The provenance layer detected all 1,000 tampered records without false positives; a hardened variant with agent signatures and external anchoring refused every forged write and exposed a history rewritten by its operator, which establishes integrity and authenticity, not truthfulness. With open LLM agents, a factorial experiment traced the change in behaviour to the wording of the reward rules rather than to the payoff.

Keywords Multi-agent systems, Explainable AI, Mechanism design, Explanation fidelity, Reputation, Blockchain provenance

1 Introduction

Decisions that used to be taken by one model are increasingly split among several specialised AI agents, each responsible for one criterion: cost, reliability, grid stability, price. Their outputs are combined by a vote or a market, and each agent attaches an explanation to its contribution so that humans can audit the result. This arrangement creates an incentive problem that single-model explainability does not face. An agent whose private objective favours one option can vote for it and attach an explanation that sounds convincing but does not reflect what its model actually computed, or that cites evidence the data do not support. Explanation methods are known to be manipulable (Slack et al. 2020; Dombrowski et al. 2019; Heo et al. 2019), and plausible but unfaithful rationalisations can be constructed on purpose (Aïvodji et al. 2019). In a multi-agent decision, nothing in the usual payoff, which rewards only the decision, makes a faithful explanation worth more to the agent than a misleading one.

Mechanism design offers a way to make truthful reporting individually rational, from peer prediction (Miller et al. 2005) to the Bayesian truth serum (Prelec 2004), and incentive mechanisms are widely used to steer self-interested learners (Zhan et al. 2021). Reputation systems give persistent weight to agents that behaved well (Jøsang et al. 2007; Kamvar et al. 2003), and blockchains can make records of AI decisions tamper-evident (Liang et al. 2017; Nassar et al. 2019). Recent work combines blockchain with incentives for multi-agent learning (Tian 2025) or LLM agent collaboration (Kumala et al. 2026), and rewards agents for faithful explanations of a model (Cheng et al. 2026). To our knowledge, however, no study puts explanation fidelity and

evidence support, measured against the agent's own model and the observed data, directly into the utility of strategic agents that jointly take a decision, and tests the result on real operational data in more than one domain.

This paper does so. The central question is whether making audited explanation quality part of each agent's payoff reduces strategic manipulation, and under which conditions it also improves the explanations themselves. Our contributions are:

1. An explanation-aware incentive for multi-agent decisions. An auditor with access to each agent's model scores every report for fidelity, evidence support and manipulation; these scores enter the agent's utility next to decision quality and contribution; rewards accumulate into a reputation that weights votes; and a provenance layer records the artefacts with integrity and, in a hardened variant, authenticity guarantees (Section 3).
2. A pre-registered evaluation on two real datasets with fully specified agents: five agents per domain with distinct private objectives and feature views, a decision-only baseline, an ablation without reputation, adversarial and collusion stress tests, tampering and forgery experiments, cross-domain comparison and sensitivity analysis (Sections 4–5).
3. Evidence on what the mechanism achieves and where it stops. It reduces manipulation in both domains; it improves explanation quality strongly where manipulation pays under the baseline and negligibly where agents are already largely honest; it does not protect the collective decision against agents that ignore the payoff; and with LLM agents the behavioural change follows the wording of the rules rather than the payoff (Section 6).

2 Related work

Explanation fidelity. SHAP attributes a prediction to features with Shapley values (Lundberg and Lee 2017), and TreeSHAP computes them exactly for tree ensembles (Lundberg et al. 2020). Whether an explanation reflects the model is its fidelity; evaluation frameworks and metrics are surveyed by Doshi-Velez and Kim (2017) and Zhou et al. (2021), and Yeh et al. (2019) formalise infidelity and sensitivity. Miller (2019) argues that explanations for people must also be selective and contrastive, which is why our agents report only their top features. We use SHAP of each agent's own model as the reference against which a reported explanation is scored.

Explanation manipulation. Post-hoc explainers can be fooled: a model can hide its reliance on a sensitive feature from LIME and SHAP (Slack et al. 2020), saliency maps can be changed arbitrarily without changing predictions (Dombrowski et al. 2019; Heo et al. 2019), and an interpretable surrogate can rationalise an unfair model (Aïvodji et al. 2019). These works study an adversary who manipulates the explainer; we study agents who misreport their own explanation, and ask whether incentives can make that unprofitable.

Incentives and reputation. Peer prediction (Miller et al. 2005) and the Bayesian truth serum (Prelec 2004) reward truthful reports without ground truth; incentive mechanisms for federated learning reward contributions of self-interested participants (Zhan et al. 2021). Reputation systems aggregate past behaviour into trust (Jøsang et al. 2007), and EigenTrust does so in peer-to-peer networks (Kamvar et al. 2003). Our mechanism has ground truth of a particular kind available, the agent's own model and the observed data, and rewards agreement with it.

Blockchain for provenance and trustworthy AI. ProvChain records data provenance on a blockchain (Liang et al. 2017); Nassar et al. (2019) argue for blockchain in explainable and trustworthy AI. Tian (2025) combine blockchain with incentive-compatible multi-agent reinforcement learning, Kumala et al. (2026) propose

blockchain-based reputation and incentives for LLM multi-agent collaboration, and Cheng et al. (2026) reward agents for faithful explanations. None of these puts the fidelity and evidence support of each agent's explanation into the utility of agents that vote on a shared decision and evaluates it on real data in two domains.

Application domains. Supplier selection in pharmaceutical supply chains is a multi-criteria problem (Forghani et al. 2018). Peer-to-peer energy trading and community markets are reviewed by Tushar et al. (2020), Sousa et al. (2019) and Tushar et al. (2018), the latter for game-theoretic approaches; the Brooklyn Microgrid is a field case (Mengelkamp et al. 2018). Our energy data come from the study of García-Muñoz et al. (2025).

3 Proposed framework

Figure 1 shows the pipeline and Figure 2 the mechanism. All parameters below were fixed in a protocol frozen before the experiments; later changes were logged as dated amendments.

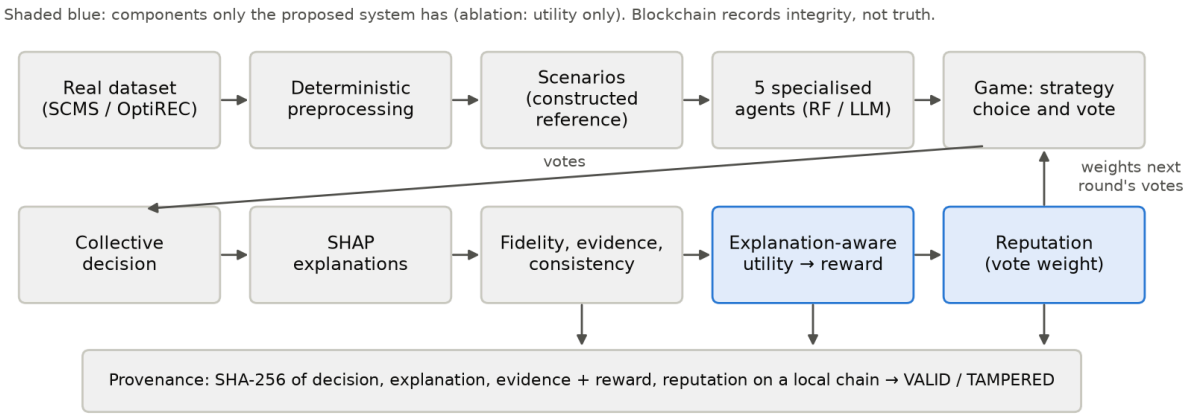


Fig. 1 Pipeline. Shaded blue: components that only the proposed system has (the ablation keeps the explanation-aware utility only). The provenance layer records integrity, not truth

3.1 One round, step by step

The framework separates three things that are easy to conflate: the *agents*, which predict and report; the *mechanism*, which audits the reports and pays rewards; and the agents' *learning rule*, which decides how an agent reports next time. The mechanism does not choose anybody's report; it only changes what each report is worth. A round proceeds as follows.

1. *Scenario.* All agents see the same real scenario: a purchase order with three candidate supplier sites, or one hour of an energy community with a battery decision.
2. *Prediction.* Each agent's own model predicts the reference decision from the agent's own features; its most probable action is the agent's *model vote*.
3. *Report.* Each agent submits a vote, a confidence, an explanation (two features and the direction of their effect) and one evidence claim per feature. The report may differ from what its model says when the agent's private objective favours another action.
4. *Audit.* The auditor compares the explanation with the SHAP attributions of the agent's model and the evidence claims with the observed data, and computes fidelity, evidence support and a manipulation score (Section 3.3).
5. *Decision, utility and reward.* The votes are combined into the collective decision; each agent's utility combines its decision quality, fidelity, evidence support, contribution and manipulation; utilities are

rescaled across agents into rewards; rewards update the reputations that weight the next votes (Section 3.4).

6. *Learning.* Each agent receives a payoff that mixes what it gains privately from the collective decision with its reward, and updates its estimate of how well its chosen reporting strategy pays (Section 3.5).

3.2 Agents, actions and reports

Each domain has five agents $N=\{1,\dots,5\}$ and a finite action set per scenario: three candidate manufacturing sites in the pharmaceutical case and {idle, charge, discharge} for a community battery in the energy case. Agent i has a private objective $u_i(a)$, a closed-form function of real or derived variables, normalised to $\tilde{u}_i(a)\in[0,1]$ over the actions of the scenario; its privately preferred action maximises $\tilde{u}_i$. Each agent also has a model, a random forest (Breiman 2001) with 100 trees of depth 6 trained on its own feature view to predict the reference decision. The private objective and the model can disagree, and that disagreement is what creates an incentive to misreport. The objectives and features of all ten agents are listed in Table 3 (Section 4).

A report is a vote, a confidence, an explanation of $k=2$ (feature, sign) pairs, and one evidence claim per explained feature stating its level (low, mid or high by training-set terciles). The strategy set has five elements. *Honest* reports the model vote, the model's probability, the top-k SHAP features with their true signs and true levels. *Strategic* switches to its privately preferred action when that gains at least 0.25 in $\tilde{u}_i$ and then inflates its confidence by 0.3, but explains truthfully. *Misattributing* and *overclaiming* vote like strategic but replace one explained feature by an unimportant one with a flipped sign, or one evidence level by a false one. *Manipulative* always votes for its preferred action with confidence 1 and reports the k least important features with flipped signs and false levels. These strategies change reports only; the observations are never altered. The collective decision is a weighted confidence vote, $\operatorname{argmax}_a \Sigma_i\, w_i\, c_i\, \mathbb{1}[v_i=a]$, with $w_i=1$ in the baseline and ablation and $w_i=Rep_i$ in the proposed system.

3.3 Auditing explanations

The auditor has white-box access to each agent's model and computes exact TreeSHAP attributions φ for the voted action (Lundberg et al. 2020). Three per-round scores follow.

Fidelity is the share of the true top-k attribution mass that the agent reported with the correct sign,

$$F_i = \frac{\sum_{j\in e_i,\ \text{sign correct}} |\varphi_j|}{\sum_{j\in \text{top-}k(|\varphi|)} |\varphi_j|},$$

with $F_i=0$ for an empty explanation and $F_i=1$ when all attributions are zero. *Evidence support* E_i is the share of the agent's claims whose stated level matches the observed value. *Manipulation* M_i is the mean of five audited components: false attribution $1-F_i$; confidence inflation $\max(0, c_i-p_i)$, where p_i is the model's probability of the vote; unsupported evidence $1-E_i$; strategic voting $\mathbb{1}[v_i\neq \text{model vote}]$; and collusion, flagged when an agent deviates from its model and another deviating agent casts the same vote. M_i is computed from reports and the audit only, never from knowledge of the strategy played. As secondary outcomes we report explanation consistency (Jaccard similarity of explanations across the five most similar scenarios), collective accuracy and manipulation success (the collective action is wrong and equals the vote of an agent that deviated from its model).

3.4 Explanation-aware utility, reward and reputation

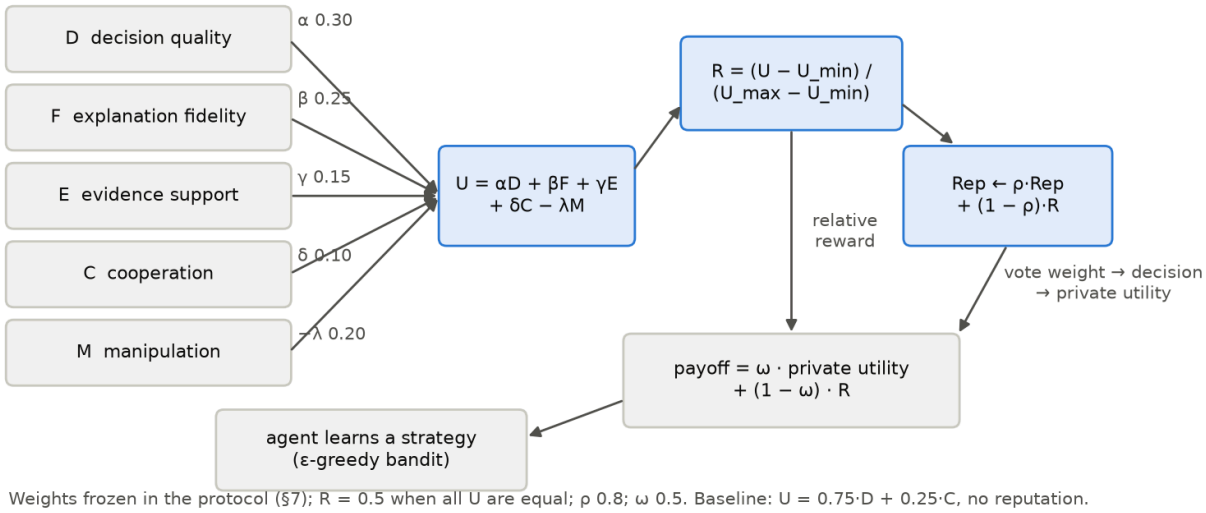


Fig. 2 The explanation-aware incentive. The utility combines five audited components; the reward is the utility rescaled across agents; reputation accumulates rewards and weights the next votes; the agent's payoff mixes its private utility of the collective action with the reward

The utility of agent i in a round is

$$U_i = \alpha D_i + \beta F_i + \gamma E_i + \delta C_i - \lambda M_i \quad (1)$$

with $D_i = \mathbb{1}[v_i$ equals the reference decision$]$ and $C_i \in \{0, 0.5, 1\}$ the agent's pivotal contribution (1 if its vote would turn a correct collective decision into a wrong one, 0 for the reverse, 0.5 otherwise). The weights $\alpha=0.30$, $\beta=0.25$, $\gamma=0.15$, $\delta=0.10$ and $\lambda=0.20$ were set a priori. The baseline uses $U_i = 0.75D_i + 0.25C_i$, the same ratio of decision to contribution with no explanation terms: under it, an explanation earns or costs nothing. The reward rescales utilities across the agents of the round, $R_i = (U_i - U_{min})/(U_{max} - U_{min})$, with $R_i = 0.5$ when all utilities are equal, so an agent is paid for doing better than the others in the same round. Reputation starts at 0.5 and is updated as $Rep_i \leftarrow \rho\, Rep_i + (1-\rho) R_i$ with $\rho = 0.8$, an effective memory of about five rounds. The mechanism therefore acts through two channels: a false report lowers the agent's reward in the round in which it is made and, through reputation, the weight of its vote in later rounds.

Table 1 Worked example: the lead-time agent in one round of the pharmaceutical proposed run (seed 0, round 354). The same round is scored with the agent's honest and manipulative report, the other agents' reports fixed

	Honest report	**Manipulative report**
Vote, confidence	site 2, 0.38	site 1, 1.00
Explanation	lead-time records raises; lead time (relative) raises	lead-time records raises; median lead time lowers
Evidence claims	lead-time records high (observed high); lead time (relative) high (observed high)	lead-time records high (observed low); median lead time high (observed low)
Fidelity F	1.00	0.00
Evidence E	1.00	0.00
Manipulation M	0.00	0.75
Decision D	0.00	0.00
Contribution C (proposed)	0.50	0.50
Utility U (proposed)	*+0.450*	*-0.100*
Reward R (proposed)	0.296	0.000
Collective decision (proposed)	site 3	site 3
Payoff π (proposed)	0.388	0.240
Contribution C (baseline)	0.00	0.00
Utility U (baseline)	*+0.000*	*+0.000*
Reward R (baseline)	0.000	0.000
Collective decision (baseline)	site 2	site 1
Payoff π (baseline)	0.000	0.500

Model vote site 2 (probabilities 0.26, 0.38, 0.36); privately preferred site 1 (normalised private utility 1.00, 0.00, 0.48); reference site 3; reputation 0.217. Payoff $\pi=\omega\tilde{u}_i(a^*)+(1-\omega)R$ with $\omega=0.5$. Baseline rows: the same reports paid by the decision-only utility with equal votes.

Worked example. Table 1 shows one real round of the pharmaceutical proposed run, chosen by a fixed rule (the first evaluation round in which a learning agent played manipulative while its preferred site differed from both its model vote and the reference). The lead-time agent's model favours site 2, its private objective favours site 1, and the reference is site 3. An honest report explains the model correctly ($F=E=1$, $M=0$) and earns a reward of 0.296. The manipulative report votes for site 1 with confidence 1 and attaches an explanation and evidence that the audit rejects ($F=E=0$, $M=0.75$); its utility becomes negative and its reward 0. Because the agent's reputation is low (0.217), its vote does not change the collective decision, so manipulation gains nothing privately and the honest report pays more (0.388 against 0.240). Paid by the decision-only baseline with equal votes, the same manipulative report would have moved the collective decision to the agent's preferred site and paid 0.500, against 0.000 for honesty. The round is an illustration; over all 1,500 agent-rounds of the evaluation window, replacing an agent's honest report by a manipulative one (the others' reports as played) paid more in 40.3% of cases under the baseline payoff and in 18.5% under the proposed one, with mean payoff differences of *+0.150* and *-0.164*.

3.5 How agents learn

Agents choose a strategy each round with a stateless ε-greedy bandit ($\varepsilon=0.1$, learning rate 0.1) on the payoff

$$\pi_i = \omega\tilde{u}_i(a^*) + (1-\omega)R_i, \quad \omega = 0.5 \tag{3}$$

where a^* is the collective action. The first term is what an agent gains privately by influencing the decision, the second how much that matters relative to the mechanism's reward. The bandit is not part of the mechanism: it is a simple model of self-interested agents that repeat what has paid. The mechanism succeeds if, under it, the

strategies that pay are the truthful ones. We do not claim that the learning dynamics reach a Nash equilibrium; instead we report, for the all-honest profile, the payoff of every unilateral deviation.

3.6 Provenance layer

We distinguish three properties a record of an agent's output can have. *Integrity*: a change after recording is detectable. *Authenticity*: the record was produced and signed by the registered agent. *Truthfulness*: the record describes what the agent's model actually computed. A blockchain can provide the first two; only the audit of Section 3.3 addresses the third.

For every agent and round, the provenance layer stores on chain the SHA-256 hashes of the canonically serialised decision, explanation and evidence, together with the reward and reputation scaled to integers ($\times 10^6$) and a block timestamp; the artefacts themselves stay off chain. A Solidity contract makes records append-only and writable only by the deploying account. Verification recomputes the hashes from the off-chain store and reports each record as valid, tampered or missing. This basic design provides integrity, but it trusts the operator who writes the records. A hardened variant, added after the main provenance results were known, removes part of that trust with three additions. First, *model registration*: each agent is bound once to a signing key and to the hash of its model identity (model, digest, decoding settings and the agent's objective). Second, *agent signatures*: the contract accepts a record only if the agent's registered key signed its identifier, artefact hashes and model hash for this chain and contract, so the operator cannot forge or replay an agent's output. Third, *anchoring*: Merkle roots (Laurie et al. 2013) of the registrations and of every 1,000 records are written to a second chain that the operator is assumed unable to rewrite, so a rebuilt history no longer matches the anchors. Rewards and reputations are computed by the mechanism and are therefore covered by the anchors but not by the agents' signatures. The hardened variant adds authenticity for the agent-produced artefacts; neither variant establishes truthfulness.

3.7 LLM agents

As an extension, we replaced the five classical agents with an open LLM, Qwen2.5-7B (Qwen et al. 2025), served locally with Ollama at temperature 0. Each LLM agent receives the scenario, the output of its model (prediction and SHAP attributions) and the reward rules, and must write its own report in the fixed format above, which is audited exactly as before.

3.8 Use of generative AI

The experiment code, analysis scripts and a first draft of this manuscript were produced with an AI coding assistant (Anthropic's Claude, used through Claude Code) under the author's direction. All reported numbers are generated by that code from the saved results; tables and figures are produced by scripts from the result files; every reference was retrieved from Crossref, DataCite or arXiv by its identifier. The author reviewed the design, the protocol amendments and the text. Qwen2.5-7B served as the experimental subject in Section 3.7, not as a writing aid.

4 Experimental design

Table 2 Case studies, scenarios and preprocessing

	Pharmaceutical	**Energy**
Source	USAID/PEPFAR SCMS delivery history (Development 2015)	OptiREC energy community (García-Muñoz 2025)
Scenario	one purchase order, three candidate sites	one hour of the community
Scenarios	1,648 (31 products, 49 sites)	8,760
Train / val / test	824 / 165 / 659 (by order date)	5,400 / 1,680 / 1,680 (week blocks)
Agents	cost, reliability, lead time, logistics, continuity	producer, consumer, market, grid, storage
Reference decision	equal-weight multi-criteria score (constructed)	daily perfect-foresight LP (constructed)
Simulated inputs	candidate set	battery: C-rate 0.5, efficiency 0.9
Derived variables	27	12

Both case studies use real operational data. Agents, scenarios and reference decisions are built from the data by deterministic, documented rules; every derived variable is a function of recorded columns, and nothing in the observations is changed by the strategies or attacks. Table 2 summarises the two datasets and Table 3 the ten agents.

Table 3 The five agents of each case study: private objective and the features its model sees

Agent	**Prefers the action with**	**Features**
Pharmaceutical supplier selection (action: one of three candidate sites)		
cost	lowest expected unit price	expected unit price, price relative to candidates, last price (relative), price variability, order size
reliability	highest past on-time delivery rate	on-time rate, on-time rate (relative), mean days late, past deliveries, late share of last five
lead time	shortest past median lead time	median lead time, lead time (relative), lead-time spread, lead-time records, requested lead time
logistics	lowest past freight cost per kg	freight per kg, freight (relative), share shipped by air, share with freight included, past deliveries to the country
continuity	most past deliveries of the product	past deliveries of the product, share of the product among candidates, days since last delivery, countries served, units of the product supplied
Peer-to-peer energy (action: charge, idle or discharge the community battery)		
producer	export when the feed-in price is above its daily median	PV availability, temperature, hour of day, feed-in price, feed-in price rank in the day
consumer	lowest cost of grid imports	community load, thermal demand, temperature, flexibility availability, import price, hour of day
market	charge when the import price is below its daily median, discharge when above	import price, feed-in price, gas price, import price rank in the day, import minus feed-in price
grid	smallest absolute exchange with the grid	net demand (load - PV), community load, PV availability, hour of day
storage	charge on community surplus, otherwise stay idle	net demand (load - PV), EVs at home, hour of day, import price rank in the day, PV availability

Each agent's private objective is normalised to *[0,1]* over the actions of a scenario; its privately preferred action is the maximiser. Each agent has its own random forest (100 trees, depth 6) trained on its own features to predict the reference decision; its argmax is the agent's model vote, and SHAP of this model is the reference for its explanation. Pharmaceutical features describe the candidate site and are computed only from deliveries completed before the order date; feature views do not overlap.

4.1 Pharmaceutical supplier selection

Data. The USAID/PEPFAR Supply Chain Management System delivery history (Development 2015) contains 10,324 shipment line items of antiretroviral drugs and HIV test kits delivered between 2006 and 2015. A line records, among other fields, the purchase order, product, manufacturing site, destination country, shipment mode, scheduled and actual delivery dates, pack price, units, weight and freight cost. We use only the 4,920

direct-from-supplier lines, because on warehouse lines the delivery performance is not the manufacturer's. A supplier is a manufacturing site; a product is a molecule or test type with its dosage and form.

Scenarios. A scenario is one purchase order for one product with three candidate sites: the three sites that most recently delivered that product before the order date, among sites with a recorded price. The data contain no competing offers, so the candidate set is an experimental construction; the site actually used is among the three in 83% of scenarios. Everything known about a candidate is computed only from deliveries completed before the order date, which prevents leakage from the order's own outcome: expected unit price (median of its five most recent prices), on-time delivery rate, median lead time, freight cost per kilogram, past deliveries of the product and related variables (27 derived variables in all). A site without lead-time or freight history receives the all-sites median (2.7% of candidate rows for lead time). The result is 1,648 scenarios over 31 products and 49 sites, split by order date into 824 training, 165 validation and 659 test scenarios.

Reference decision. Each of five criteria (cost, delivery reliability, lead time, freight cost per kilogram, supply continuity) is min–max normalised over the three candidates, and the reference is the candidate with the highest equal-weight sum, in the spirit of multi-criteria supplier selection (Forghani et al. 2018). Equal weights are used because the source gives none. This is a constructed reference, not a record of good decisions: where the site actually used is a candidate, the reference picks it in 35% of scenarios, close to chance (33%). Dropping one criterion leaves the reference unchanged in 68–74% of scenarios and doubling one in 72–82%.

Agents. Each agent represents one criterion and prefers the candidate that is best on it (Table 3); its model sees five features of that criterion only. Each agent's preferred site coincides with the reference in 46–52% of scenarios, so every agent often has a private reason to vote against the reference. This conflict is what makes manipulation profitable under a decision-only payoff.

4.2 Peer-to-peer energy

Data. The OptiREC dataset (García-Muñoz 2025; García-Muñoz et al. 2025) contains one year of hourly operational data of a 55-user energy community under peer-to-peer trading and flexibility markets: per-user load profiles and generation capacities, PV availability, import and feed-in prices, gas price, temperature, thermal demand, flexibility availability, electric vehicle presence and battery capacity. It has no missing values; no outlier was removed. Derived variables (12) are deterministic functions of the recorded columns, for example community load, PV output, net demand (load minus PV), and the rank of the hour's price within its day.

Scenarios and reference decision. A scenario is one hour (8,760 in all), and the decision is the action of an aggregate community battery: charge, idle or discharge. The data record no storage dispatch, so the reference is the solution of a daily perfect-foresight linear programme that minimises import cost minus feed-in revenue subject to the battery's energy balance. The battery capacity is taken from the data; its C-rate (0.5), round-trip efficiency (0.9), state-of-charge window (10–90%) and daily start and end state (50%) are simulated assumptions, because the dataset does not provide them. The reference is therefore constructed as well, and decision accuracy means agreement with this programme, not with observed behaviour. Splits use whole weeks in five-week cycles (three for training, one for validation, one for testing; 5,400, 1,680 and 1,680 hours), which prevents temporal leakage.

Agents. The producer, consumer, market, grid and storage agents represent the perspectives of a P2P community (Table 3). Their objectives are driven by prices and surplus, as is the reference programme, so they conflict with the reference less often than the pharmaceutical criteria do; the results in Section 5.1 show what this implies.

4.3 Systems compared

Three systems are run on identical scenarios and seeds (paired design): the *baseline* (game and explanations, decision-only utility, equal votes), the *ablation* (explanation-aware utility, equal votes) and the *proposed* system (explanation-aware utility and reputation-weighted votes). Each system is run with seeds 0–9: 600 rounds with evaluation on the last 300 in pharma, 1,000 rounds and the last 500 in energy.

4.4 Hypotheses and statistical analysis

The protocol pre-registered six hypotheses, each comparing the proposed system with the baseline:

- **H1** the proposed system improves explanation fidelity;
- **H2** it improves evidence support;
- **H3** it reduces manipulation;
- **H4** it is more robust to an increasing share of malicious agents;
- **H5** the provenance layer detects modification of recorded artefacts;
- **H6** H1–H3 give the same verdict in both domains.

H1–H3 form the confirmatory family, tested per domain with a paired *t*-test or, when Shapiro–Wilk rejects normality, a Wilcoxon signed-rank test, with Holm correction (Holm 1979) across the three. The pre-registered decision rule calls an outcome *improved* if the Holm-adjusted $p<0.05$ in the hypothesised direction and the relative change $(proposed-baseline)/|baseline|$ is at least 5%, *worse* if significant in the opposite direction, and *negligible* otherwise. The 5% threshold separates practical from merely statistical significance: a significant change smaller than 5% is reported as negligible. H4 is tested as a stress test: malicious shares of 0–40% (malicious agents always play manipulative, whatever the payoff), comparing the mean accuracy degradation over 10–40% between systems with a paired test; collusion uses two colluding agents and three honest ones. H5 is descriptive. H6 holds for an outcome when both domains give the same verdict; domains are never pooled. The comparison of the proposed system with the ablation, which isolates the contribution of reputation, is a secondary analysis. The sensitivity analysis varies one parameter at a time: $\beta \in \{0,0.1,0.25,0.4\}$, $\rho \in \{0.5,0.8,0.95\}$ and $\omega \in \{0.25,0.5,0.75\}$. The LLM agents (Section 3.7) answer a separate question, whether such agents respond to the payoff at all; they were run for 200 rounds with seeds 0–4 and are reported apart from the confirmatory family.

5 Results

5.1 Explanation quality and manipulation

Table 4 Baseline vs proposed, classical agents (*n=10* paired seeds; mean ± SD). Verdicts: pre-registered rule (Holm *p<0.05* in the hypothesised direction and *|rel.|≥ 5%*)

Domain	Metric	Baseline	Proposed	*Δ* [95% CI]	Rel.	*p* (Holm)	Verdict
pharma	Fidelity *F*	0.510 ± 0.077	0.936 ± 0.020	+0.426 [+0.370, +0.482]	+83.5%	*<0.0001*	improved
	Evidence support *E*	0.522 ± 0.070	0.917 ± 0.023	+0.395 [+0.345, +0.444]	+75.7%	*<0.0001*	improved
	Manipulation *M*	0.392 ± 0.038	0.119 ± 0.015	-0.273 [-0.302, -0.243]	-69.7%	*<0.0001*	improved
	Collective accuracy	0.578 ± 0.045	0.606 ± 0.033	+0.029 [-0.018, +0.076]	+5.0%	0.2021[a]	–
	Manipulation success	0.398 ± 0.057	0.263 ± 0.032	-0.135 [-0.185, -0.085]	-33.9%	0.0002[a]	–
energy	Fidelity *F*	0.936 ± 0.024	0.970 ± 0.004	+0.034 [+0.021, +0.050]	+3.6%	0.0039	negligible
	Evidence support *E*	0.926 ± 0.031	0.966 ± 0.004	+0.040 [+0.017, +0.063]	+4.3%	0.0039	negligible
	Manipulation *M*	0.066 ± 0.017	0.036 ± 0.005	-0.029 [-0.043, -0.016]	-44.4%	0.0026	improved
	Collective accuracy	0.744 ± 0.019	0.754 ± 0.013	+0.009 [-0.000, +0.019]	+1.2%	0.0541[a]	–
	Manipulation success	0.050 ± 0.017	0.025 ± 0.012	-0.025 [-0.037, -0.012]	-49.4%	0.0015[a]	–

[a]Secondary outcome, unadjusted *p*, no verdict.

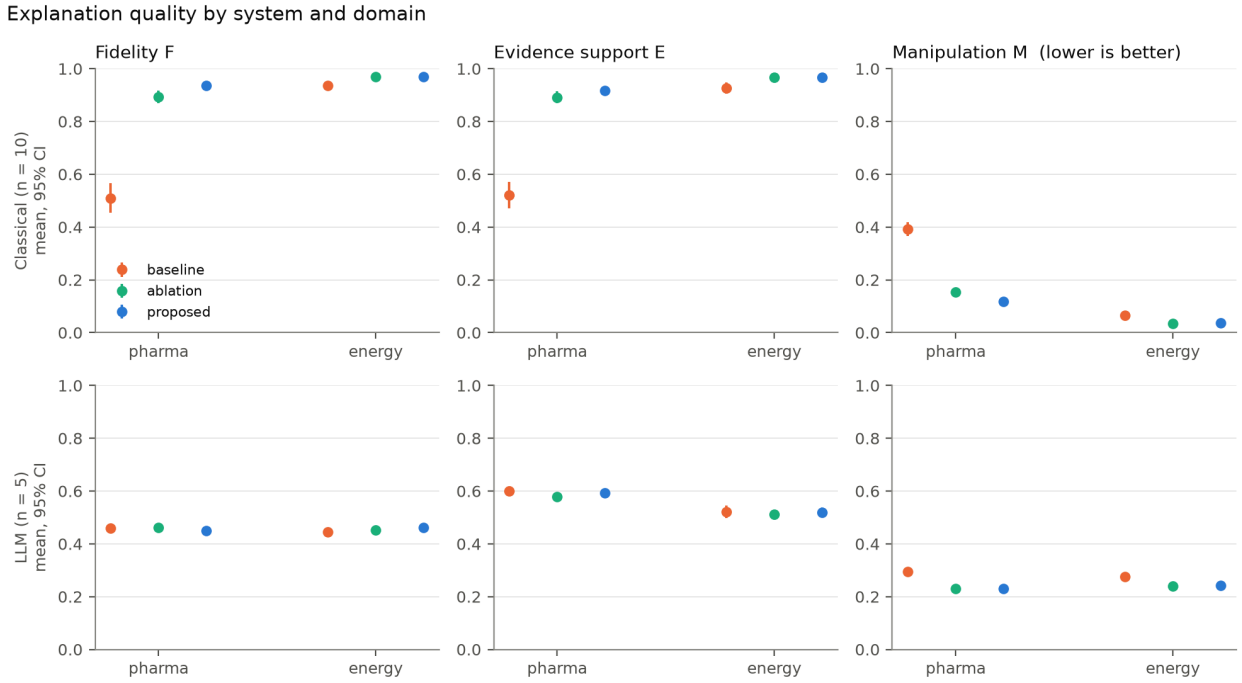


Fig. 3 Fidelity, evidence support and manipulation by system and domain, for classical (top) and LLM agents (bottom); points are means, bars 95% confidence intervals

Table 4 and Figure 3 show the main comparison. In the pharmaceutical case the incentive raised fidelity from 0.510 to 0.936 (*+83.5%*) and evidence support from 0.522 to 0.917 (*+75.7%*), and cut manipulation from 0.392 to 0.119 (*-69.7%*); all three are improved under the decision rule (Holm *p<0.0001*). In energy, manipulation fell from 0.066 to 0.036 (*-44.4%*, improved), while fidelity (*+3.6%*) and evidence support (*+4.3%*) rose significantly but below the 5% threshold, so both are negligible. The energy baseline already reached *F=0.936* and *E=0.926*, leaving little room to improve; we report this as context and do not change the verdict. Collective accuracy did not change detectably in either domain, while manipulation success fell by a third in pharma and by half in energy (secondary outcomes). Explanation consistency rose by 28.6% and 8.6%.

Why the two domains differ is visible in what the agents learned. Under the baseline, the pharmaceutical agents played honest in only 9.5% of evaluation rounds and manipulative in 41.1%; under the proposed system, 41.4% and 3.0%. The energy agents were honest in 77.5% of rounds already under the baseline (87.6% under the

proposed system), because their objectives rarely pay them to steer the battery away from the reference. Where there is little manipulation to remove, there is little explanation quality to gain. The relative changes should be read with this in mind: a relative change measures improvement over the baseline's level, so the same mechanism produces a large relative gain where the baseline is poor and a small one where it is already near its ceiling.

The unilateral-deviation analysis qualifies these results. In energy, honesty is the best response for all five agents in every system, so the baseline is already close to honest. In the pharmaceutical case, honesty is the best response for no agent under the baseline and for two of five under the proposed system; the other three gain most from strategic voting with a truthful explanation. The incentive therefore removes the profit of false explanations without making the all-honest profile an equilibrium.

5.2 Ablation: incentive versus reputation

Table 5 Incentive ablation (incentive without reputation) vs baseline, and the descriptive increment of reputation (proposed - ablation)

Domain	Metric	Rel.	*p* (Holm)	Verdict	Reputation increment
pharma	Fidelity *F*	+74.9%	*<0.0001*	improved	+0.044
	Evidence support *E*	+70.9%	*<0.0001*	improved	+0.025
	Manipulation *M*	-61.0%	*<0.0001*	improved	-0.034
energy	Fidelity *F*	+3.5%	0.0037	negligible	+0.001
	Evidence support *E*	+4.4%	0.0037	negligible	-0.001
	Manipulation *M*	-46.2%	0.0013	improved	+0.001

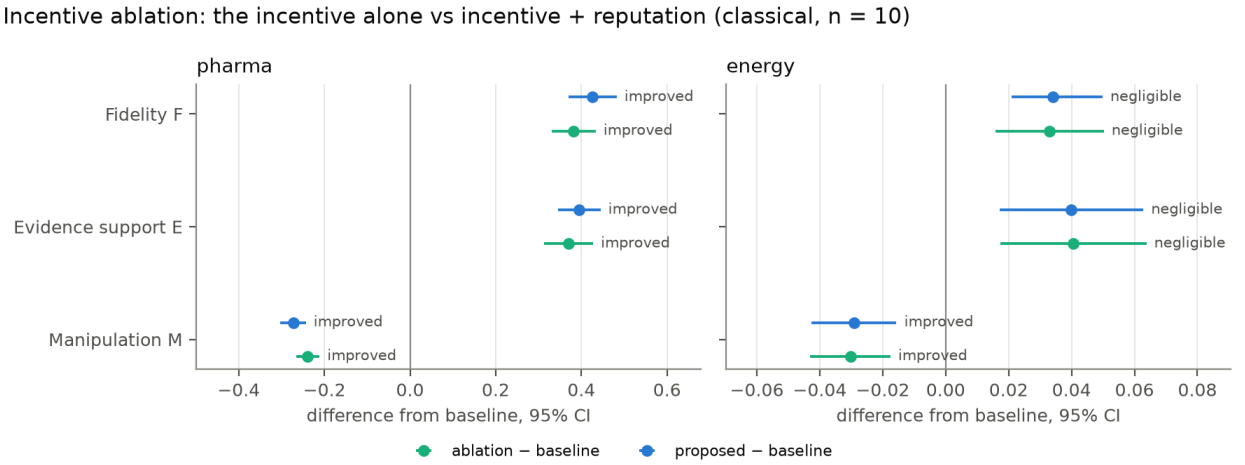


Fig. 4 Difference from the baseline for the ablation (incentive only) and the proposed system (incentive and reputation), with 95% confidence intervals and verdicts

The explanation-aware utility alone produces most of the effect (Table 5, Figure 4): the ablation reaches the same verdicts as the proposed system in both domains, with relative changes in pharma of *+74.9%* (fidelity), *+70.9%* (evidence) and *-61.0%* (manipulation). Reputation-weighted voting adds a small further gain in pharma (fidelity *+0.044*, manipulation *-0.034*) and none in energy; this comparison is descriptive.

5.3 Adversarial stress test: a boundary of the mechanism (H4)

Table 6 Robustness. H4: mean accuracy degradation over 10–40% malicious agents, proposed - baseline (negative = proposed degrades less; paired *t*-test). Collusion: change in manipulation success (baseline / proposed) and in the colluders' reward under the proposed system when two agents collude (unadjusted)

Domain	H4 *Δ* [95% CI]	*p*	Verdict	Collusion: manip. success	Colluders' reward
pharma	+0.039 [-0.009, +0.087]	0.1021	not supported	+0.380 / +0.126	-0.458
energy	-0.020 [-0.043, +0.002]	0.0684	not supported	+0.155 / +0.047	-0.501

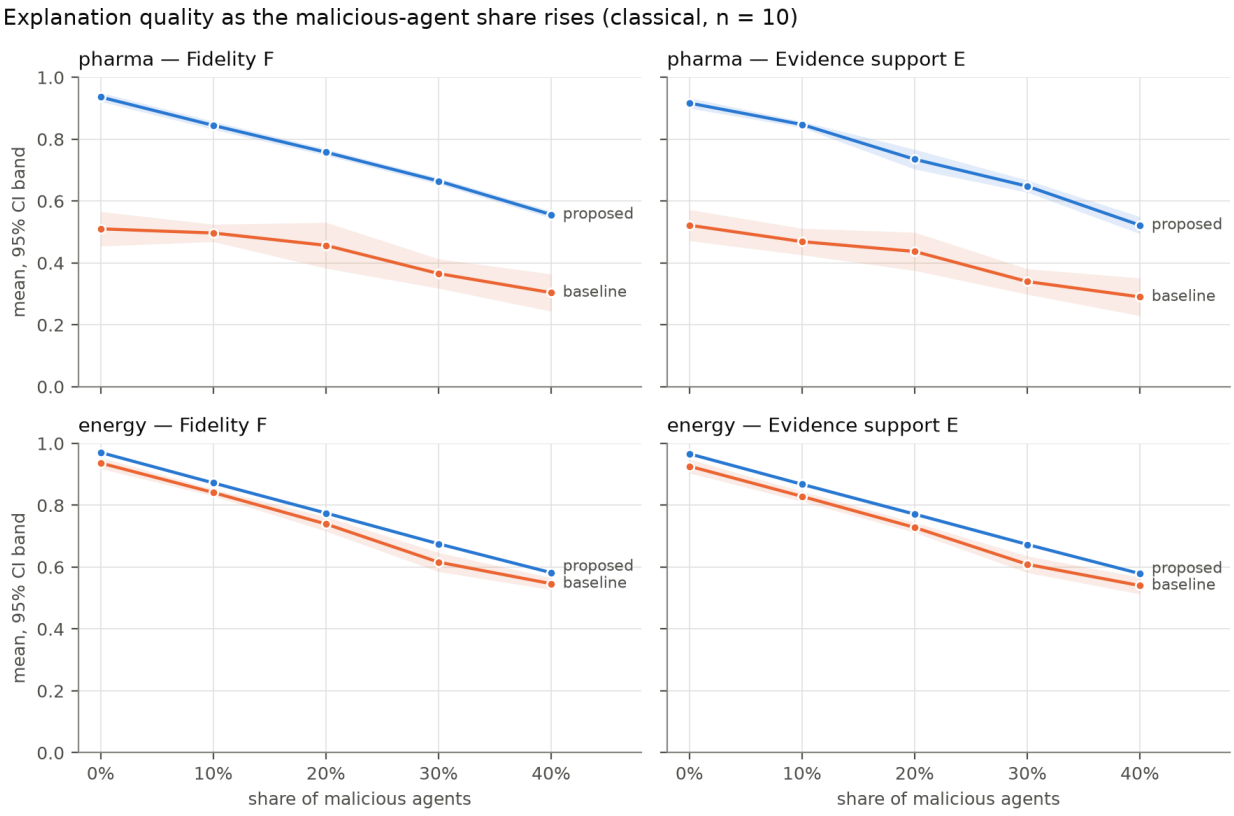


Fig. 5 Fidelity and evidence support as the share of malicious agents rises (means and 95% confidence bands)

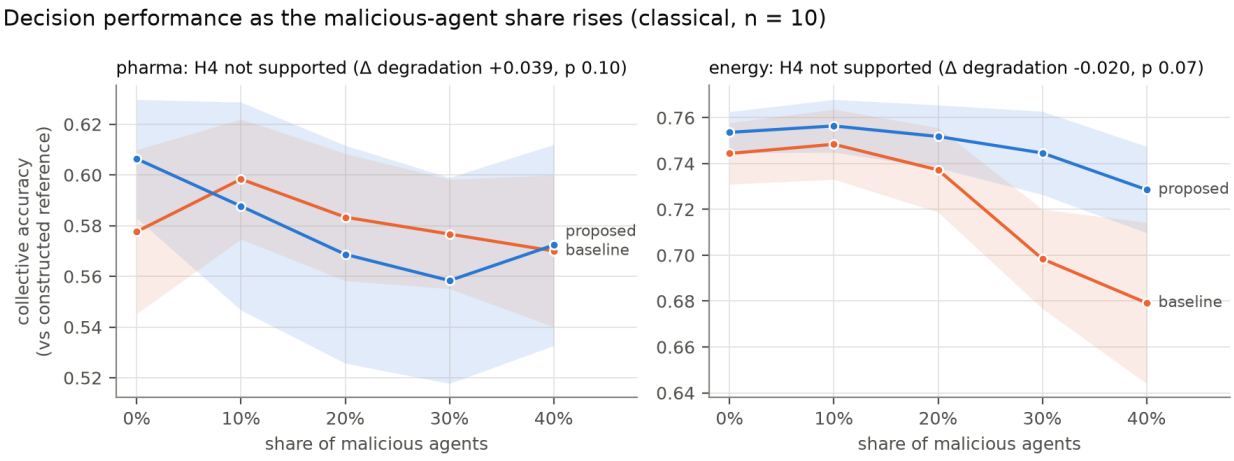


Fig. 6 Collective accuracy as the share of malicious agents rises

This experiment asks a different question from H1–H3. There, agents choose their reports and the mechanism changes what pays; here, a growing share of agents manipulates every round regardless of the payoff. H4 is not supported (Table 6). The proposed system's accuracy did not degrade less than the baseline's as malicious agents rose from 10 to 40%: the mean difference in degradation was *+0.039* (*p=0.10*) in pharma and *-0.020*

($p=0.068$) in energy (Figure 6). The proposed system keeps clearly higher fidelity and evidence support than the baseline at every malicious share, but both fall roughly linearly as the share grows (Figure 5): the audit scores what malicious agents report, and the mechanism cannot make them report differently. Under collusion, the mechanism takes the reward away from the colluders (*-0.458* in pharma, *-0.501* in energy) and the honest agents' payoff rises; collusion still raises manipulation success, but by less than under the baseline (*+0.126* against *+0.380* in pharma; *+0.047* against *+0.155* in energy).

5.4 Provenance: integrity and authenticity (H5)

Table 7 Tamper detection on the provenance layer (H5)

Group	Flagged TAMPERED	Meaning
decision	200 / 200	detected
explanation	200 / 200	detected
evidence	200 / 200	detected
reward	200 / 200	detected
reputation	200 / 200	detected
untouched controls	0 / 1,000	false positives
re-serialised controls	0 / 1,000	false positives
all other records	0 / 27,800	false positives
below 10^{-6} resolution	0 / 200	not detectable by design

Local node, 30,000 records. Batches of 100: 95,912 gas per record, 8.5 ms verification per record; 128 bytes on-chain per record.

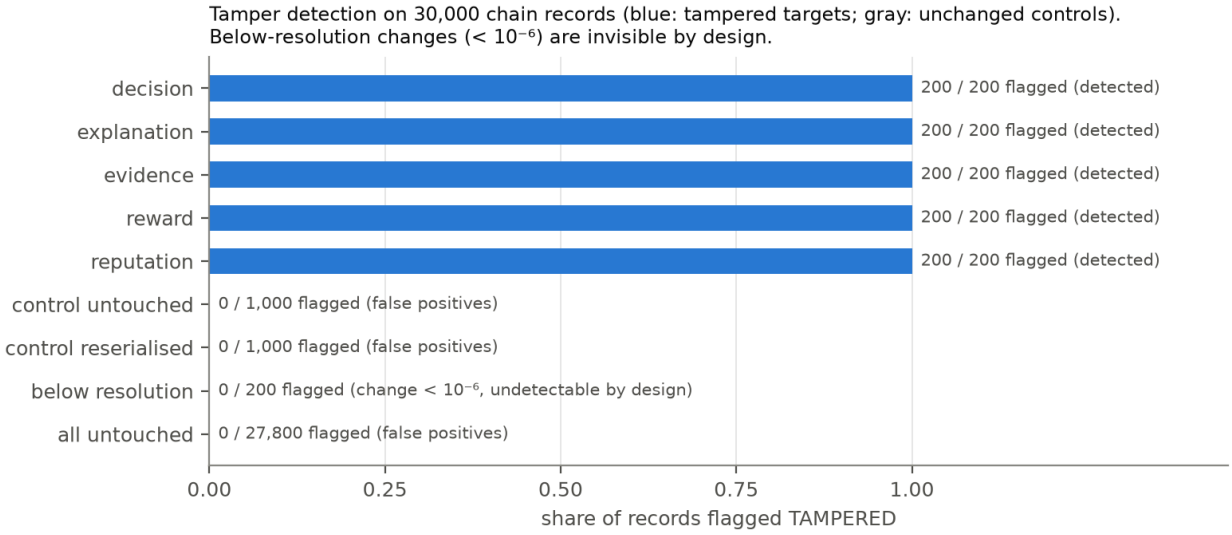


Fig. 7 Share of records flagged as tampered, by target and control group

The provenance layer recorded 30,000 agent-round records from the LLM main runs of both domains on a local Ethereum-compatible node. Of 1,000 records altered off chain, 200 in each of decision, explanation, evidence, reward and reputation, all were detected; no false positive occurred in 2,000 untouched or re-serialised controls or in the 27,800 other records (Table 7, Figure 7). Changes smaller than the 10^{-6} storage resolution of rewards and reputation are invisible by design. The contract refused all overwrites and writes from another account. In batches of 100, a record cost about 96,000 gas; local-node latency excludes consensus and network delay and will differ on a production network. H5 is supported in the sense of integrity: the layer shows that records were not changed, not that they were true.

Table 8 Hardened provenance: agent signatures, model registration and anchoring (amendment 9, descriptive)

Check	Result
Forged or replayed writes (refused by the contract)	
content changed after signing	0 / 100 accepted
signed with another agent's key	0 / 100 accepted
signed with an unregistered key	0 / 100 accepted
replayed under another record id	0 / 100 accepted
replayed from another deployment	0 / 100 accepted
substituted model	0 / 100 accepted
sent by a non-owner account	0 / 100 accepted
correctly signed (control)	100 / 100 accepted
History rebuilt by the operator	
altered records, amendment-5 check	0 / 200 detected
registrations against anchor	10 / 10 differ
altered records against anchor	200 / 200 detected
unaltered records against anchor	0 / 29,800 flagged
Honest run	
records, registrations, anchors	0 tampered, 0 + 0 mismatches

Local nodes, 30,000 records; the anchor chain is a second local node standing in for an external chain. Batches of 100: 126,005 gas per record (amendment 5: 95,912); anchoring 113 gas per record; inclusion proof 320 bytes.

The hardened variant was tested on the same 30,000 records, with the anchor chain simulated by a second local node (Table 8). The contract refused every forged or replayed write: content changed after signing, signatures by another agent's or an unregistered key, signatures replayed under another record identifier or from another deployment, and substituted models (100 attempts each), while all 100 correctly signed control records were accepted. When the operator rebuilt the entire history with 200 altered records, per-record verification found nothing, as expected when chain and off-chain store are both replaced; the registrations exposed the operator's keys for all ten agents, and the anchors identified exactly the 200 altered records, whichever field was changed, with no unaltered record flagged. The hardening raised the cost to about 126,000 gas per record in batches of 100 (from 96,000), plus 113 gas per record for anchoring. These results are descriptive and rest on the assumption that the anchor chain cannot be rewritten.

5.5 Cross-domain comparison and sensitivity

Table 9 Cross-domain comparison (H6: same verdict in both domains) and robustness of the classical verdicts across β, ρ and ω

Agents	Outcome	Pharmaceutical	Energy	Same verdict
Classical	Fidelity F	+83.5% (improved)	+3.6% (negligible)	no
	Evidence support E	+75.7% (improved)	+4.3% (negligible)	no
	Manipulation M	-69.7% (improved)	-44.4% (improved)	yes
LLM	Fidelity F	-1.9% (negligible)	+3.4% (negligible)	yes
	Evidence support E	-1.0% (negligible)	-0.3% (negligible)	yes
	Manipulation M	-22.0% (improved)	-12.6% (improved)	yes
Sensitivity (pharma)	verdicts changed	none (all values of β, ρ, ω)		
Sensitivity (energy)	verdicts changed	ω: fidelity, evidence		

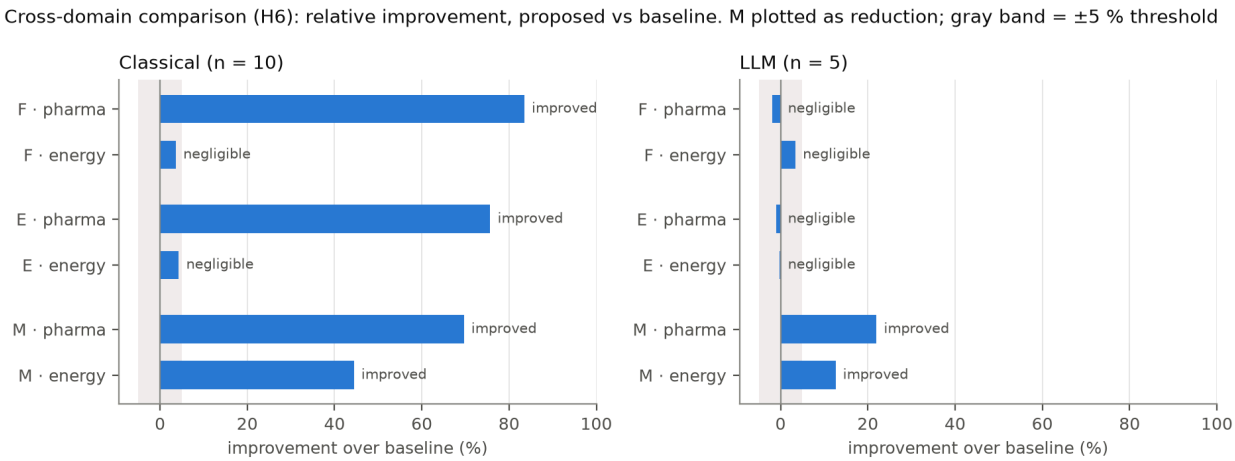


Fig. 8 Relative improvement of the proposed system over the baseline in each domain; manipulation is plotted as a reduction; the grey band marks the 5% threshold

With classical agents, both domains give the same verdict only for manipulation, so H6 holds for one of three outcomes (Table 9, Figure 8). The pharmaceutical verdicts are unchanged at every value of β, ρ and ω. In energy they are unchanged across β and ρ, but fidelity and evidence support change from negligible to improved at $\omega=0.75$ (*+27.8%* and *+28.6%*): when steering the decision is worth more, the baseline's explanations deteriorate and the mechanism's advantage appears. In pharma the fidelity gain persists at $\beta=0$ (*+74.7%*), so it is produced by the evidence and manipulation terms together and should not be attributed to the fidelity weight alone.

5.6 LLM agents

Table 10 LLM agents: 2×2 wording × payoff experiment on mean stated confidence ($n=5$ seeds; Holm over the three effects; added after the main analysis)

Domain	Effect	Δ [95% CI]	p (Holm)	Verdict
pharma	wording	-0.299 [-0.306, -0.291]	*<0.0001*	lower
	payoff	+0.001 [+0.000, +0.002]	0.0971	no detectable effect
	interaction	-0.004 [-0.008, -0.000]	0.0971	not detected
energy	wording	-0.164 [-0.167, -0.161]	*<0.0001*	lower
	payoff	-0.000 [-0.002, +0.002]	0.9884	no detectable effect
	interaction	+0.006 [+0.004, +0.009]	0.0046	detected

With LLM agents (Table 9), manipulation fell in both domains (*-22.0%* and *-12.6%*, Holm $p<0.0001$) while fidelity and evidence support were negligible; fidelity stayed between 0.45 and 0.46 in every system; sample replies show the model misreading the signs and levels in its own tool output. The fall in manipulation came mostly from lower stated confidence: the confidence-inflation component accounts for *-0.060* of the *-0.065* change in pharma and *-0.029* of *-0.035* in energy. To find its cause we added, after the main analysis, a 2×2 experiment that crosses the wording of the reward rules in the prompt with the payoff actually applied (Table 10). The wording lowered confidence by 0.299 in pharma and 0.164 in energy, whereas no payoff effect was detected in either domain; the energy run, specified before it was carried out, reproduced the pharmaceutical pattern, with a small wording-by-payoff interaction (*+0.006*) in energy only. For these agents the behavioural change reflects what they are told, not what they are paid.

6 Discussion

What the mechanism does. Putting audited explanation quality into the utility reliably reduced manipulation: in both domains, with classical and LLM agents, in the ablation without reputation and across the sensitivity range. It also removed the advantage of false explanations: in the pharmaceutical case the agents that still deviate do so with truthful explanations. Whether explanation quality itself improves depends on how much the baseline leaves to gain. In pharma, where unilateral manipulation pays under a decision-only utility, the gains were large; in energy, where honesty was already a best response, they were significant but small, and they became large only when the private stake ω rose.

What it does not do. The mechanism did not make the collective decision more robust to a growing share of agents that always manipulate. It scores reports; it cannot change the votes of agents that ignore the payoff, and reputation reacts only after several rounds. Defences against persistent attackers need another instrument, such as exclusion thresholds or robust aggregation. With LLM agents the improvement cannot be credited to the payoff: a prompt that states the rules produces the same change. Designers of incentive mechanisms for LLM agents should therefore test whether agents respond to payoffs at all before attributing behaviour to them.

Implications. The results indicate when an explanation-aware incentive is worth deploying: where agents' private objectives conflict with the collective goal, so that manipulation pays under a decision-only payoff. There it removes the profit of false explanations and shifts agents towards truthful reports; where agents are already aligned, it adds little. It requires an auditor with access to the agents' models, and it has to be combined with a separate defence for agents that do not respond to incentives at all. For provenance, designers should state which of integrity, authenticity and truthfulness a ledger provides; ours provides the first two, and only the audit speaks to the third.

Security scope. The evaluated threats are dishonest agents with valid identities (false attribution, inflated confidence, unsupported evidence, strategic voting, collusion of two), changes to recorded artefacts, and forged, replayed or rewritten records. Out of scope are attackers who adapt to the utility, Sybil identities and whitewashing (a new identity starts at reputation 0.5), poisoned source data, models built to mislead SHAP, a declared model that differs from the one actually run, stolen agent keys, and network-level attacks; all components ran in one process on one machine. The provenance layer is a research prototype, not a system hardened against infrastructure attacks.

Limitations. Both reference decisions are constructed, and decision accuracy means agreement with them, not with real choices. The pharmaceutical data cover one programme's antiretroviral and HIV-test deliveries, and the candidate sets are inferred. The energy battery parameters are simulated. Fidelity is measured against SHAP of the agent's own model and therefore needs white-box access; it inherits the limits of SHAP and says nothing about whether the model itself is right. The terciles that define evidence levels and the incentive weights were set a priori; other choices would give other numbers. Reputation creates path dependence. Inference rests on ten seeds (five for LLM agents) on one dataset per domain. The LLM results come from one 7-billion-parameter model, and the factorial experiment was added after the main analysis, as was the hardened provenance layer, whose anchor chain was simulated by a second local node. Overhead was measured on local nodes and will differ on production networks.

7 Conclusion

We proposed an explanation-aware incentive for multi-agent AI decisions, in which audited explanation fidelity, evidence support and manipulation enter each agent's utility, reputation weights votes and a blockchain records the artefacts, and evaluated it under a pre-registered protocol on real pharmaceutical and energy data. The incentive reduced strategic manipulation in both domains and removed the profit of false explanations; it improved explanation quality strongly only where the baseline left room; it did not increase robustness to

malicious agents; and the provenance layer detected every tampered record and, with agent signatures and anchoring, refused every forged write and exposed a history rebuilt by its operator. For LLM agents, the effect followed the wording of the rules rather than the payoff. Future work should test the mechanism with black-box fidelity estimates, real decision outcomes, larger and more diverse LLMs, and defences that act on persistent attackers.

Declarations

Funding. [To be completed on the title page.]

Competing interests. [To be completed on the title page.]

Data availability. The pharmaceutical data are the public USAID SCMS delivery history (Development 2015); the energy data are available from Mendeley Data (García-Muñoz 2025). Processed scenarios are regenerated deterministically by the code.

Code availability. The code, configuration files, frozen protocol with its amendments, per-experiment summaries and the scripts that generate every table and figure will be released in a public repository on acceptance; an anonymised copy is available to reviewers.

Author contributions. [To be completed on the title page.]

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